

Removing Toxic Plastic Additives from the Human Body

A biomedical program to block the adverse health effects of chronic chemical exposure

Rotem Gura Sadovsky, PhD · Biotech Science Leader and VC · rgura@csb.co.jp

Program Goal. Develop biomedical solutions that act within the human body to reduce the toxicity of ingested microplastics. Microplastics continuously enter our bodies through food¹, drinks² and air³ starting the day we are born⁴⁻⁶. These microplastics contain additives that harm fertility⁷, increase the risk of diabetes⁸, and may interfere with brain development in young children⁹. Exposure to these harmful additives is ubiquitous: they have been consistently detected in more than 90% of people¹⁰. Removing these chemicals from our bodies will avert a chronic deterioration in the health of 100 million people in Japan and billions globally.

Current Approaches and Their Limitations. The gold standard for managing chemical exposure is banning or restricting the use of chemicals through environmental regulation. Japan exemplifies this approach, with a government commitment through the National Action Plan for Marine Plastic Litter¹¹, and more recently through funding a comprehensive research program on pollution and health, with the goal of informing additional environmental regulation¹². However, environmental regulation is inherently limited in protecting human health, because it addresses new sources of chemicals while neglecting those that already exist in the human body and that will persist in the environment for centuries.

Meanwhile, the healthcare system offers no complement. Tests that measure plastic additive levels in the body are expensive and do not reflect the true chronic exposure unless taken many times. More importantly, the results of these tests are not actionable, as no approved treatments exist. Billions of exposed people have no medical pathway to reduce their exposure to plastic additives.

Technical Rationale. Three convergent developments make a biomedical approach timely. First, *human epidemiological data at scale are imminent*. Cohorts launched in the 2010s, including the Japan Environmental and Children Study (JECS)¹³ which includes 100,000 newborns, will soon publish the first large-scale human data on prenatal chemical exposure and reproductive endpoints. In addition to this longitudinal study, new interventional studies show that changing the pattern of plastic exposure is reflected in measured biomarkers^{15,16}. These advances lay the data foundation required for biomedical innovation.

Second, *gut-resident technological proofs of concept exist*. Engineered bacteria^{17,18}, hydrogels^{19,20}, enzymes²¹, nucleic acid aptamers²² and gut-anchored nanostructures²³ all represent technological components that can be assembled into products for blocking, degrading or neutralizing plastic additives. It is important that chemical removal will take place in the human gut, because the gut serves as a gateway, where systemic absorption into the body is either allowed or prevented.

Third, *a biomedical research and innovation ecosystem around plastics is emerging*. In April 2026, the US government agency ARPA-H launched STOMP²⁴, a \$144 million program for creating scalable clinical tests for microplastic levels and understanding their effects on human health. This funding has attracted the interest of hundreds of research groups, effectively

seeding a research ecosystem that brings biomedical approaches to plastic exposure. This emerging ecosystem will be a natural source of transition funding for the products of this proposed program in Japan.

The program proposed here will leverage these developments to build biomedical products for capturing or degrading plastic additives in the body. The program will fund multiple groups (“performers”) in parallel, each developing differentiated technology, to increase the chances of technical success. The program will focus on solutions active in the gut, but will consider solutions active in the lungs if they demonstrate sufficient technical readiness. The success of this program will be defined as proof-of-concept products that achieve 80% reduction in selected plastic additives, bisphenols and phthalates, in animal models within three years.

Performers selected for the program will be evaluated against three profiles corresponding to the program's technical areas. Groups developing binding or degradation technologies must demonstrate direct expertise in relevant modalities, including enzyme engineering, biomaterials, aptamers, or microbial engineering. Groups developing gut-deployment platforms will have experience in scaffolds, hydrogels, or engineered bacteria designed to anchor to or persist within the intestinal lumen or mucus layer. Testing in model systems will require groups with demonstrated capacity to expose human organoids or animal models to chemicals and to measure biomolecular responses, including hormonal, metabolic, or inflammatory biomarkers. Priority will be given to Japan-based groups, but international groups will be considered.

Impact on Japanese Society. Japan's population is highly exposed to plastics. Japan tracks with other developed nations in ubiquitous use of polyester clothing, bottled drinks, and food packaging, and it ranks highest per capita in seafood consumption, especially tuna, which contains high amounts of plastic additives²⁵ from marine environments polluted with microplastics, mostly from degraded car tires and paint.

At the same time, Japan is suffering from decreasing fertility rates and increasing chronic disease rates. Japan's fertility rate has been dropping for decades and in recent years reached 1.15, the lowest ever recorded²⁶. Biological infertility remains a significant factor, where assisted fertility is responsible for 11% of live births^{27,28}, and is cumulatively successful in only about 50% of women²⁹. In addition, Inflammatory Bowel Disease rates have been increasing for decades, including a 40% increase since 2015³⁰. Diabetes rates have also increased, by 100% in Japanese men and by 50% in Japanese women between the 1990s and the 2010s³¹.

The extent to which plastic additives contribute to these health trends is partially known, and is an active area of research³². The ubiquitous plastic additives bisphenols and phthalates lower sperm quality⁷, disrupt metabolic balance linked to diabetes⁸, and are especially harmful to IBD patients, who are more sensitive to exposure due to their compromised gut barrier³³.

If this program successfully blocks the effects of plastic additives in the body, it will address a biological contributor to infertility and confer direct health benefits in other conditions that may be linked to plastic additives. Beyond health benefits, successful products developed by this program will catalyze new company creation opportunities and support existing startups in gut-based therapeutics, positioning Japan as an innovation leader in the new field of biomedicine for pollutant exposure, and in the growing field of gut health.

The bigger picture contribution of this program globally is unblocking the path for adoption of biomedical solutions to chronic chemical exposure. Once treatments exist, public awareness will skyrocket, human data will become plentiful, and next generation products will be funded and

developed. Ultimately, chemical exposure will be measured and addressed as part of standard clinical care for every person. This new standard will lead not only to better healthcare, but also to knowledge of which of the myriad chemicals we synthesize harm our health, and fuel material innovation, improving human and environmental health at the same time. If Japan builds expertise in biomedical treatments for chemical exposure, it would be well-positioned to lead in this new field.

Risks and Mitigation.

1) *Developed solutions may bind or degrade native human biomolecules or deplete essential gut nutrients alongside target pollutants.* These pollutants structurally mimic endogenous molecules, making biochemical specificity a technical risk. Mitigation: funding diverse technological approaches in parallel to increase the probability that some of the solutions achieve the required substrate specificity.

2) *Regulators may not approve clinical trials if levels of plastic additives are insufficiently linked to clinical indications.* Mitigation: companies spun out of this program will be equipped to enter the consumer product space, where the regulatory bar is lower than for approved drugs, enabling market entry and data generation ahead of a future clinical pathway for next generation products. To enable that path, products will need to show sufficient preclinical effectiveness and outstanding safety profile.

3) *Developed products may not attract sufficient private investment due to uncertain consumer demand.* Mitigation: transition funding is already in view: ARPA-H's STOMP program is expected to open its second funding phase in 2029, targeting microplastic removal solutions. Other government and philanthropic programs will likely emerge as the field is growing rapidly.

Cost, structure and deliverables. The proposed program will cost ¥3B, tranching across three years. The program will operate in an ARPA-like model with three technical areas: developing capture and degradation solutions for plastic additives, anchoring these solutions to the human gut, and testing the solutions in model systems. Performer awards will average ¥400M each. The program will deliver proof-of-concept technologies safe and effective in animal models within three years. Additional outputs include Intellectual Property filings, new company spinouts, and ties with the emerging global innovation ecosystem in biomedical solutions for microplastics.

Timeline and milestones.

Timeline (months)	Test system	Success criterion	Performers meeting milestone
M9	Solution in buffer	Removes ≥ 3 priority chemicals	5 of 6–8
M18	Simulated gut fluid ³⁴	Specific blocking demonstrated	4
M24	Gut organoids	Substrate specificity and no toxicity	3
M36	Animal models	80% removal of ≥ 5 priority chemicals, no toxicity	2

References

1. Kadam-Czapska, K., Knez, E. & Grembecka, M. Food and human safety: the impact of microplastics. *Crit. Rev. Food Sci. Nutr.* **64**, 3502–3521 (2024).
2. Gambino, I., Bagordo, F., Grassi, T., Panico, A. & Donno, A. D. Occurrence of Microplastics in Tap and Bottled Water: Current Knowledge. *Int. J. Environ. Res. Public Health* **19**, (2022).
3. Vianello, A., Jensen, R. L., Liu, L. & Vollertsen, J. Simulating human exposure to indoor airborne microplastics using a Breathing Thermal Manikin. *Sci. Rep.* **9**, 8670 (2019).
4. Caba-Flores, M. D., Martínez-Valenzuela, C., Cárdenas-Tueme, M. & Camacho-Morales, A. Micro problems with macro consequences: accumulation of persistent organic pollutants and microplastics in human breast milk and in human milk substitutes. *Environ. Sci. Pollut. Res.* **30**, 95139–95154 (2023).
5. Minatoya, M. *et al.* Cord Blood Bisphenol A Levels and Reproductive and Thyroid Hormone Levels of Neonates: The Hokkaido Study on Environment and Children's Health. *Epidemiology* **28 Suppl 1**, S3–S9 (2017).
6. Yin, Z. *et al.* First identification of microplastics in umbilical cord blood and their direct target proteins: A pioneering discovery. *Ecotoxicol. Environ. Saf.* **307**, 119405 (2025).
7. Wang, H. *et al.* Do phthalates and their metabolites cause poor semen quality? A systematic review and meta-analysis of epidemiological studies on risk of decline in sperm quality. *Environ. Sci. Pollut. Res.* **30**, 34214–34228 (2023).
8. Martínez-Pinna, J. *et al.* Endocrine disruptors in plastics alter β -cell physiology and increase the risk of diabetes mellitus. *Am. J. Physiol. Endocrinol. Metab.* **324**, E488–E505 (2023).
9. Yu, Z.-X. *et al.* Risk of Neurodevelopmental Disorders in Preschool Children Associated with the Longitudinal Trajectory of Phthalates during Pregnancy: Potential Mechanisms Based on Metabonomics of Cord Blood. *Environ. Sci. Technol.* **59**, 16180–16192 (2025).
10. Vandenberg, L. N. *et al.* Urinary, Circulating, and Tissue Biomonitoring Studies Indicate Widespread Exposure to Bisphenol A. *Environ. Health Perspect.* **118**, 1055–1070 (2010).
11. On the Formulation of the Action Plan for Marine Plastic Litter Countermeasures (in Japanese). 環境省 <https://www.env.go.jp/press/106865.html>.
12. Program Overview | Comprehensive Environmental Risk Research Program. https://www.nies.go.jp/risk_health/program2023/program-e.html.
13. Study Documents | About the Study | Japan Environment and Children's Study. <https://www.env.go.jp/chemi/ceh/en/about/advanced/document.html>.
14. Joubert, B. R. *et al.* Powering Research through Innovative Methods for Mixtures in Epidemiology (PRIME) Program: Novel and Expanded Statistical Methods. *Int. J. Environ. Res. Public Health* **19**, 1378 (2022).
15. Harray, A. J. *et al.* Low-plastic diet and urinary levels of plastic-associated phthalates and bisphenols: the randomized controlled PERTH Trial. *Nat. Med.* **32**, 1871–1883 (2026).
16. Hua, J. *et al.* Targeting Plastic Exposure in Infertile Couples: A Pilot Intervention Study. *Toxics* **14**, (2026).
17. Lindell, A. E. *et al.* Human gut bacteria bioaccumulate per- and polyfluoroalkyl substances. *Nat. Microbiol.* **10**, 1630–1647 (2025).
18. Lee, J. *et al.* Efficient biosorption of nanoplastics by food-derived lactic acid bacterium. *Bioresour. Technol.* **447**, 134234 (2026).
19. Dutta, S., Misra, A. & Bose, S. Polyoxometalate nanocluster-infused triple IPN hydrogels for excellent microplastic removal from contaminated water: detection, photodegradation, and upcycling. *Nanoscale* **16**, 5188–5205 (2024).
20. Kim, A.-R., Mitra, S. K. & Zhao, B. Unlocking Passive Collection of Microplastics in Coral Reefs by Adhesion Measurements. *ACS EST Water* **4**, 5653–5659 (2024).

21. Zurier, H. S. & Goddard, J. M. PETase Engineering for Enhanced Degradation of Microplastic Fibers in Simulated Wastewater Sludge Processing Conditions. *ACS EST Water* **3**, 2210–2218 (2023).
22. Selection of Plastic-Binding DNA Aptamers for Microplastics Detection - Zandieh - 2025 - *Angewandte Chemie International Edition* - Wiley Online Library.
<https://onlinelibrary.wiley.com/doi/full/10.1002/anie.202421438>.
23. Shahgaldian, P., CORRERO-SHAHGALDIAN, M. R., Cumbo, A. & CORVINI, P. F.-X. Biocatalytical composition. (2016).
24. STOMP | ARPA-H. <https://arpa-h.gov/explore-funding/programs/stomp> (2026).
25. Wang, M.-H. *et al.* Microplastic ingestion and phthalate esters bioaccumulation in yellowfin tuna (*Thunnus albacares*): tissue-specific distribution across gills, stomach, and muscles. *Mar. Pollut. Bull.* **219**, 118257 (2025).
26. World Bank Open Data. *World Bank Open Data* <https://data.worldbank.org>.
27. 2025 White Paper on Health, Labour and Welfare: Appendix | Ministry of Health, Labour and Welfare. https://www.mhlw.go.jp/wp/hakusyo/kousei/25-2/dl/01_en.pdf.
28. Katagiri, Y. *et al.* Assisted Reproductive Technology in Japan: A Summary Report for 2023 by the Committee on Professional Scientific Conduct and Clinical Ethics of the Japan Society of Obstetrics and Gynecology. *Reprod. Med. Biol.* **25**, e70008 (2025).
29. Jwa, S. C. *et al.* Cumulative live birth rate according to the number of receiving governmental subsidies for assisted reproductive technology in Saitama Prefecture, Japan: A retrospective study using individual data for governmental subsidies. *Reprod. Med. Biol.* **20**, 451–459 (2021).
30. Tsutsui, A. *et al.* Nationwide estimates of patient numbers and prevalence rates of ulcerative colitis and Crohn's disease in Japan in 2023. *J. Gastroenterol.* **60**, 1513–1522 (2025).
31. Fauzi, M., Kartiko-Sari, I. & Poudyal, H. Trends of Dietary Intakes and Metabolic Diseases in Japanese Adults: Assessment of National Health Promotion Policy and National Health and Nutrition Survey 1995–2019. *J. Clin. Med.* **11**, (2022).
32. Peng, C. *et al.* Atmospheric aging of tire rubber antioxidants forms complex mixtures that trigger inflammation in human macrophages. *Environ. Int.* **212**, 110307 (2026).
33. Chen, X. *et al.* Adverse health effects of emerging contaminants on inflammatory bowel disease. *Front. Public Health* **11**, 1140786 (2023).
34. Roberts, B. & Thaarup, I. Simulated Media for Mimicking the Human Environment In Vitro. *Apmis* **133**, e70024 (2025).